

Pressure is only part of diagnosis

Glaucoma is an optic neuropathy; pressure must be interpreted alongside the nerve, visual field and angle.



Normal optic nerve head

Glaucomatous cupping

The paired fundus images compare a normal optic nerve head with glaucomatous cupping. The right-hand image has a larger pale cup and reduced neuroretinal rim. The embedded arrows and labels belong to the supplied image; image position does not establish eye laterality.

Combine structure and function

Assess tonometry, disc appearance, visual fields and gonioscopy together. POAG has an open angle, characteristic damage and no secondary cause. Ocular hypertension is raised pressure without detectable damage. Normal tension glaucoma has characteristic damage despite consistently normal-range pressure. A single reading may mislead because IOP fluctuates; record timing and consider corneal thickness. Normal central acuity does not exclude clinically important field loss. Progressive rim or field damage remains important even when the measured pressure seems satisfactory.

Recognise and break an acute attack

A painful red eye with vomiting, a cloudy cornea and a fixed mid-dilated pupil suggests acute angle closure.

Pupillary block sequence

Iris-lens contact impedes aqueous passage through the pupil



Posterior chamber pressure exceeds anterior chamber pressure



Peripheral iris bows forward and obstructs the angle



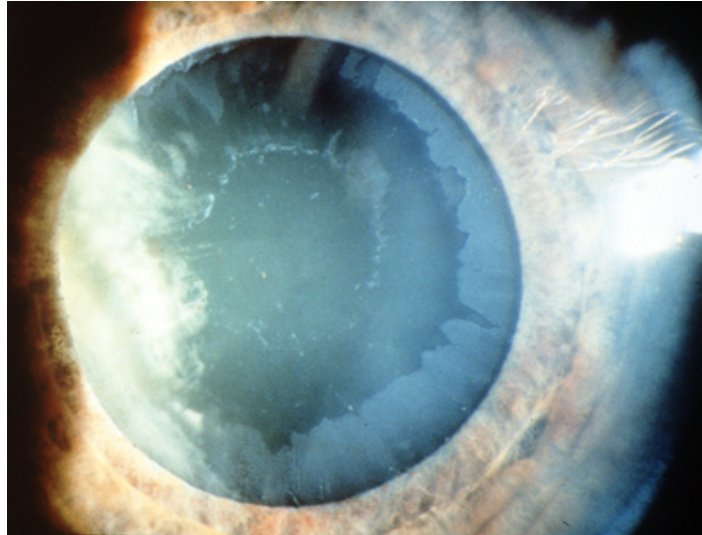
Pressure rises sharply; urgent medical lowering is required

Urgency and definitive care

A shallow chamber, haloes and markedly elevated pressure support the diagnosis. Begin urgent ophthalmic pressure-lowering treatment and manage pain and nausea; pupil dilation can worsen block. Once the attack is controlled and the cornea permits laser, peripheral iridotomy provides an alternative posterior-to-anterior chamber route. Assess the fellow eye for prophylactic treatment. Angle closure can also be intermittent or asymptomatic, so the absence of a dramatic attack does not exclude disease.

Find secondary causes

An open angle does not imply primary disease; anterior segment findings may reveal the obstructing material.



Whitish pseudoexfoliative material is visible on the anterior lens capsule. The same process can deposit material in the trabecular meshwork and increase resistance to aqueous outflow while the angle remains open.

Identify the mechanism

Pigment dispersion releases iris pigment that obstructs the trabecular meshwork. Phacolytic disease involves leakage of lens material from a mature cataract; phacomorphic disease instead involves lens enlargement contributing to closure. Inflammation and steroid exposure can also raise trabecular resistance. With iris neovascularisation, treat pressure and the underlying retinal drive using PRP and anti-VEGF as appropriate. Established fibrovascular synechial closure will not simply reverse after anti-VEGF. Look for deposits, inflammation, cataract and new vessels rather than labelling every high-pressure open angle POAG.

Choose and monitor treatment

The aim is to preserve useful visual function by slowing damage, with treatment adapted to risk and progression.

Drug clues and cautions

Class	Useful distinction
Beta blocker	Reduces production; asthma, bradycardia and heart block matter
Prostaglandin analogue	Increases uveoscleral outflow; iris darkening and lash growth
Alpha-2 agonist	Production and outflow effects; avoid in young children
Carbonic anhydrase inhibitor	Reduces production; systemic adverse effects require review

Judge response by damage

Set a target pressure in light of existing damage, baseline pressure and patient factors. If reproducible damage progresses despite reaching that target, lower the target and reassess treatment. Different drug classes may be combined; laser or surgery can be considered when adequate control is not achieved. Follow-up includes the disc and field, not only a pressure number. Check adherence and systemic contraindications before assuming that a drop is suitable for every patient.